Simulation of Spherical Bubble Dynamics

Numerical Simulation Using Python 2.7



Anant Seth

2nd Year BTech Student

INTRODUCTION

Rayleigh Plesset equation, which is a simplified version of Navier Stokes equation can be used to predict the bubble growth and collapse in an infinite liquid, under the assumptions of spherical symmetry, incompressible liquid and negligible gas inertia inside the liquid.Below is the Rayleigh Plesset equation,

$$\frac{p_V(T_\infty) - p_\infty(t)}{\rho_L} + \frac{p_{Go}}{\rho_L}\left(\frac{R_o}{R}\right)^{3k} = R\ddot{R} + \frac{3}{2}(\dot{R})^2 + \frac{4\nu_L\dot{R}}{R} + \frac{2S}{\rho_L R} \quad \text{...eq(1)}$$

With k=1(taken from Cavitation-and-bubble-dynamics by Christopher Earls Brennen [1])

But due to assumption of the spherical symmetry by the Rayleigh Plesset equation , it cannot predict any jetting phenomena or any other type of asymmetry in the bubble development arising from the local flow/boundary configuration/forcing terms.

In the simulation two equations are used first the Rayleigh Plesset equation and second the modified form with the integration term.Below is the eq with the integral term.

$$\frac{Pv - P\infty}{\rho} + \frac{Pg\circ}{\rho}\left(\frac{R\circ}{R}\right)^3 = R\ddot{R} + \frac{3}{2}\dot{R}^2 + \frac{4v\dot{R}}{R} + \frac{2s}{\rho R} - \frac{12}{\rho}\int_0^t \frac{\dot{R(x)}R(x)}{R(x)^3 - R(t)^3} e^{-\frac{(t-x)}{\lambda}} \log\frac{R(x)}{R(t)} dx$$

..eq (2)

Where *lambda=dynamic viscosity /Elastic Modulus*

In both the simulation the same set of values are taken :

the liquid taken is 'Water'

1. ρ (water liquid density)=998.2kg/m3
2. pv is the vapour pressure. = 0 Pa
3. p(inf) is the pressure at the bubble level, including the hydrostatic pressure,
4. p(inf) =10350Pa.
5. pg0 (initial bubble pressure) = 10^8 Pa
6. σ (surface tension)= 0.072N/m.
7. μ (dynamic viscosity of water) = 1.01.10-3Pa.s

Simulation 1:

The Rayleigh Plesset equation (eq 1) was used in the first simulation.In order to solve the Second order differential equation ,Simultaneous Runge Kutta fourth order numerical method was used.

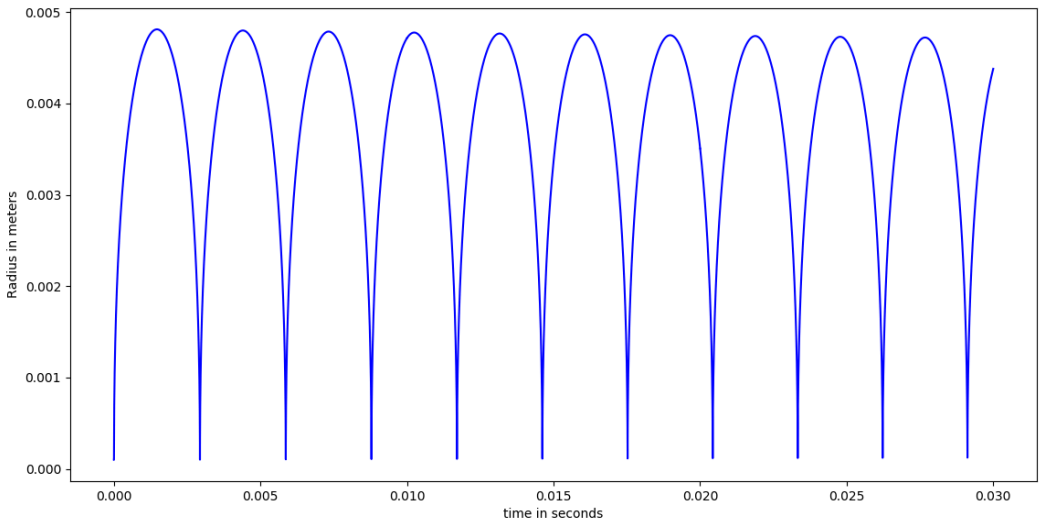
Observation 1 :

With

1. R is the bubble radius, = 0.1 mm
2. Step size in Runge Kutta method ‘h’ = 10^-7 sec

Result:

Graph :Radius(in meters) vs Time(sec)



Referring the paper : Simulation of bubble expansion and collapse in the vicinity of a free surface[2]

P. Koukouvinis , M. Gavaises , O. Supponen, M. Farhat City University London, Northampton Square, London EC1V 0HB, United Kingdom 5 EPFL-LMH, Avenue de Cour 33 Bis, CH-1007 Lausanne

The observation as per the given paper are :

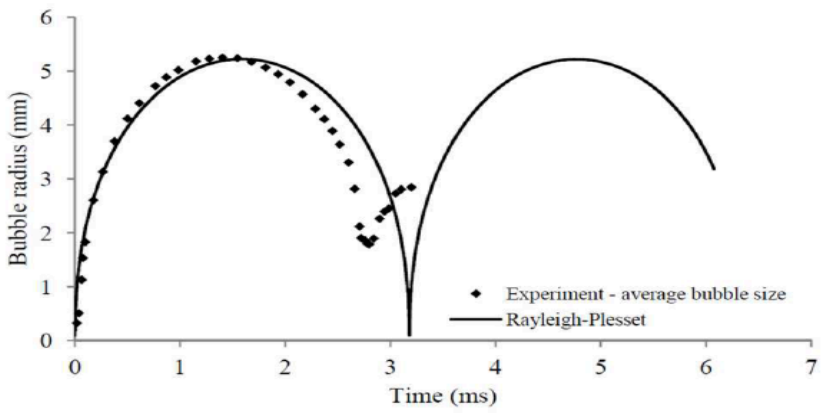


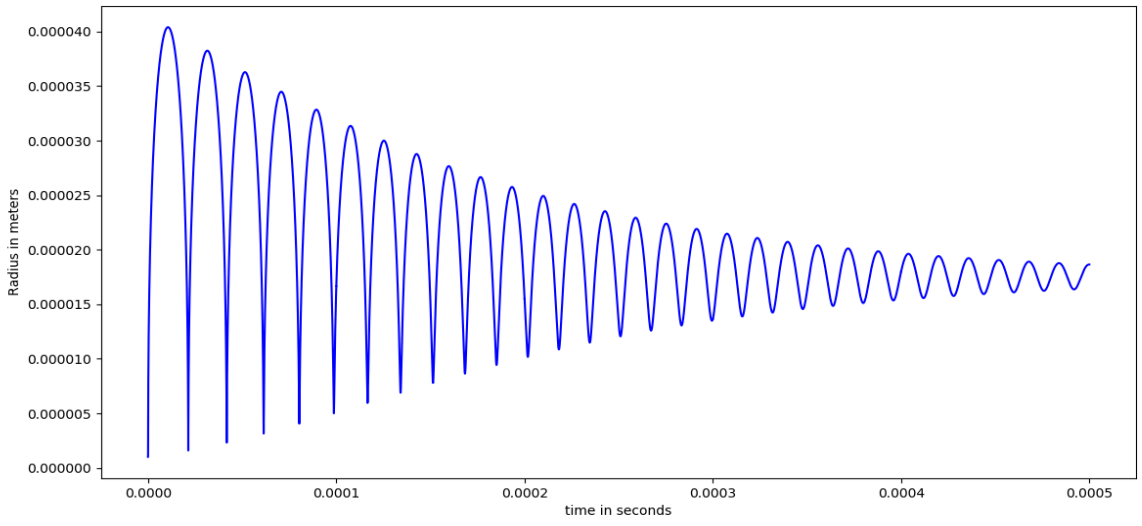
Figure 6. Time evolution of the experimental bubble size and comparison with the Rayleigh-Plesset solution for R_0=0.1mm and p_{g0}=1000bar.

Observation 2 :

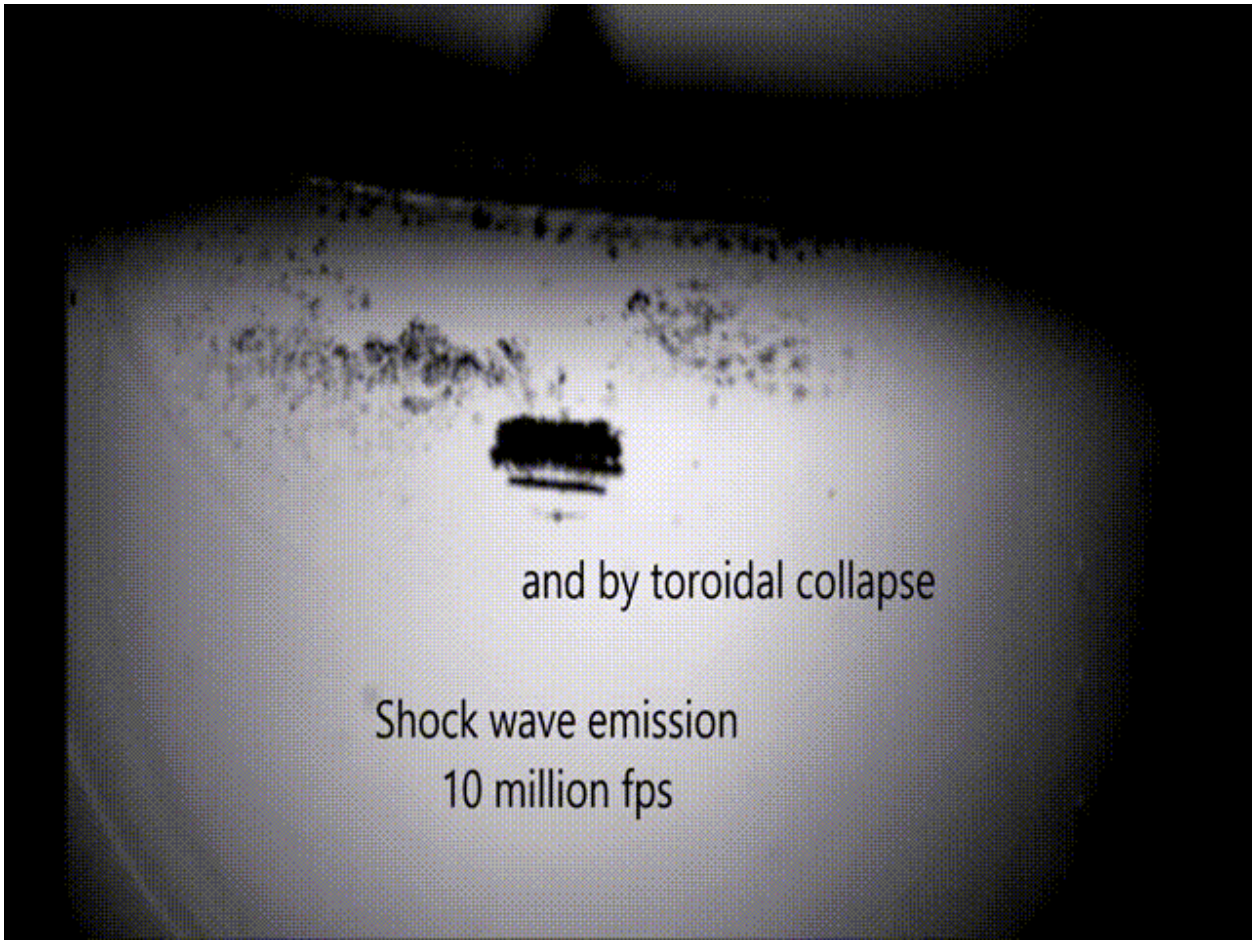
1 . R is the bubble radius, = 0.1 micro meter

2. Step size in Runge Kutta method 'h' = 10^-9 sec

Result :



The video simulation of bubble collapse (Reference https://www.youtube.com/watch?v=ZaB5B3pec-A)



The video clearly shows the jetting phenomena .

RESULTS:

Oscillating graph between Radius of the bubble and Time is obtained in both the graphs in accordance with the paper on *Simulation of bubble expansion and collapse in the vicinity of a free surface[2]* as well as the above video simulation .

Simulation 2:

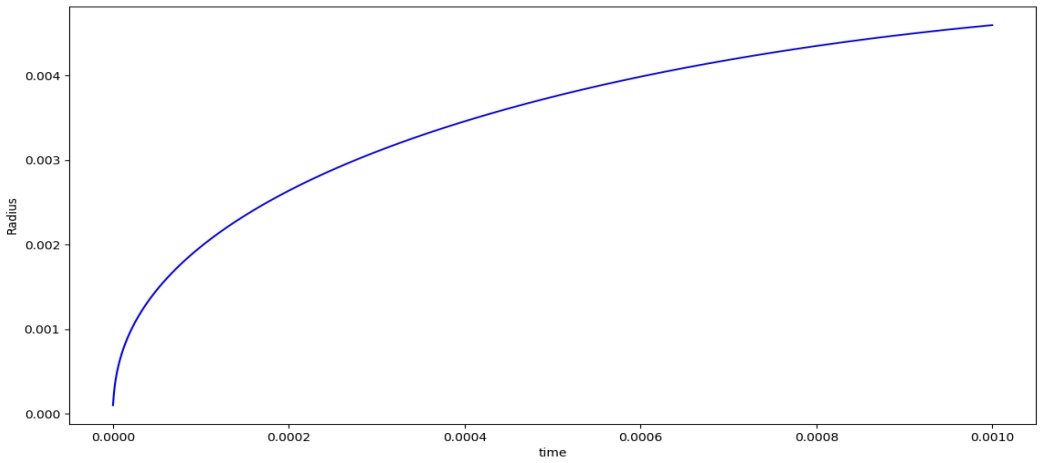
In the simulation both eq 1 and eq 2 was used.In order to solve the equation, first the(eq 1) second order differential equation is solved using Runge Kutta Method .Then with the help of Cubic Spline Interpolation method , R(t) and R'(t) function were generated and with the help of Simpson's method of integration the integral term was calculated and again the the Integro -Differential equation was solved using same Runge Kutta method.

Due to large computation of the program used in simulation ,the iteration was set to smaller range value.

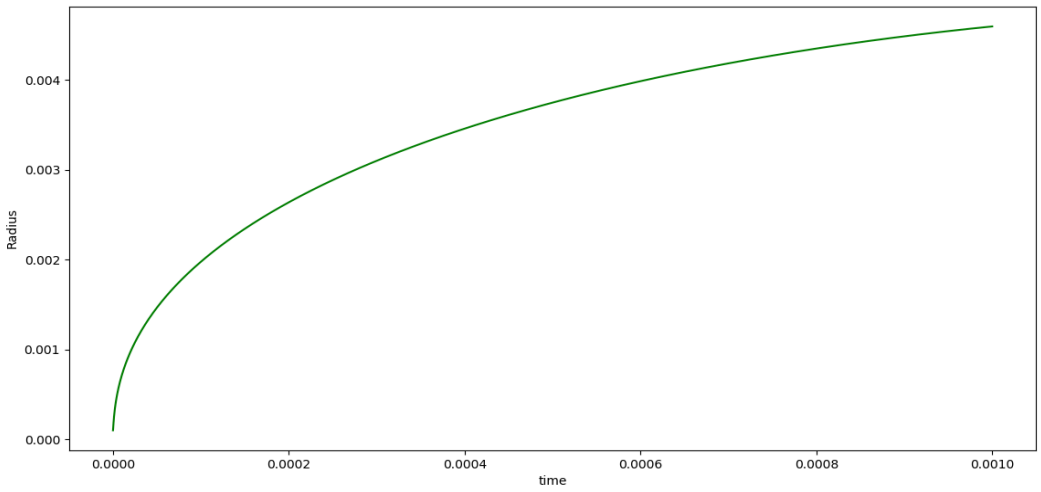
Observation :

1. R is the bubble radius, = 0.1 mm
2. Step size in Runge Kutta method 'h' = 10^-7 sec

First graph without the integration term



With integration term



Result:

It can be observed that both the graphs are similar due to the fact the exponential part present in the Integro - Differential Equation becomes quiet small , very near to zero and thus the value of the function in the Integral part becomes very near to zero and therefore does not affects the overall equation leading to similar graph.

The graph in Simulation 2 would be similar to the graph in Simulation 1 if program used in simulation 2 iterated for a long time.

Reference:

1.)Inner World of Collapsing bubble (video):https://www.youtube.com/watch?v=ZaB5B3pec-A

2.)Cavitation and Bubble Dynamics - Caltech Authors

3.)Simulation of bubble expansion and collapse in the vicinity of a free ...

4.)Theoretical study of bubble dynamics in purely viscous fluids